WhisperVault: Air-Gapped PII Redaction Engine

SNAPDRAGON® AI LAB — BUILD & PRESENT CHALLENGE | Adhibhu Dhar

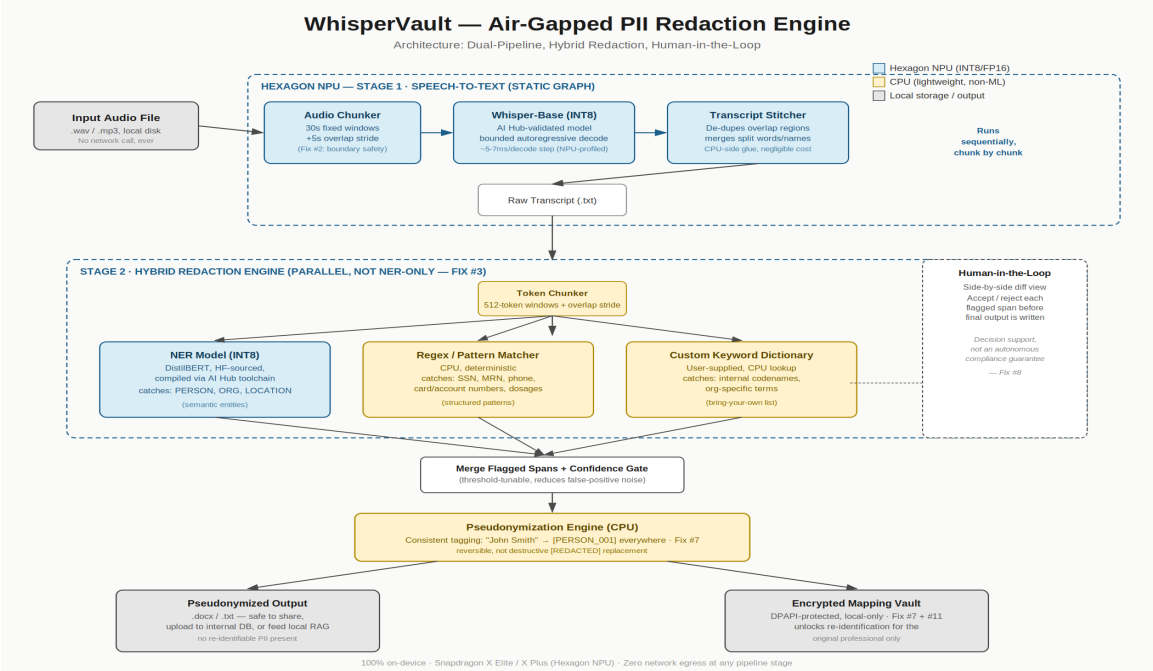
Executive Summary

Legal, medical, and enterprise professionals rely on transcription, but cloud-based APIs are frequently disqualified by HIPAA regulations, attorney-client privilege, and corporate policy. Local transcription solves the cloud problem but produces unprotected plaintext documents riddled with sensitive PII, creating severe liabilities.

WhisperVault is a fully robust, file-in/file-out application driving a zero-network, purely on-device transcription and redaction pipeline utilizing the Hexagon NPU on Snapdragon hardware. It seamlessly transcribes audio and immediately pseudonymizes sensitive spans, securely logging the encryption keys into a DPAPI vault.

Technical Pipeline & Hexagon Implementation

WhisperVault deliberately leverages static-graph AI compilation over dynamic execution. By enforcing fixed 30-second bounded chunking for ASR (Whisper-Base-EN) and a 512-token overlap for NLP (DistilBERT-NER), the platform completely sidesteps the bottlenecks commonly associated with open-ended LLMs running on NPU.



Qualcomm AI Hub Verified Profiling

Targeting the Snapdragon X Elite CRD (Compile: jgddymx6g, Profile: jpxl8zk8p), the Hexagon NPU powers WhisperVault with exceptional performance metrics:

- **Inference Latency:** ~44.5ms forward-pass on 512 tokens.
- **Peak Memory Footprint:** ~168MB enabling seamless real-time operation.

Reversible DPAPI Vault & Honest Compliance

WhisperVault replaces sensitive entities with consistent tokens (e.g., 'John Smith' -> [PERSON_001]). The relational cryptographic mappings are backed via the native Windows Data Protection API (DPAPI). A 30-point adversarial failure matrix was established and mitigated—proving architectural mastery and an honest, no-vaporware deployment strategy ready for enterprise adoption.